

Unit 6 – Inference for Categorical Data: Proportions
Notes 1 – Confidence Intervals for Population Proportions

Confidence Intervals: The Basics

- If we want to estimate the value of a _____ from a population, we can take a sample and use our sample _____ to estimate p .
- BUT we know that selecting different samples will yield different values of $\hat{p}$, and we will therefore get a different estimate of p each time.
- Because of this, instead of giving a _____ estimate for p , we give an _____ of plausible values for the population parameter, using $\hat{p}$
- A confidence interval is an interval of _____ for an unknown parameter value
- For a population proportion p , we can estimate its value using $\hat{p}$ _____ $\pm$ _____ as our _____ and constructing a margin of error around it
- A margin of error is constructed with a degree of confidence and the standard deviation of the sampling distribution
- The confidence level describes how “confident” we are the interval captures p

Example Forms:

Confidence Levels

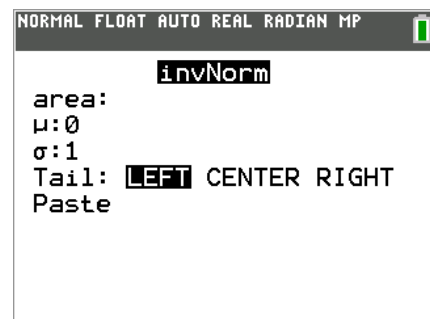
- The confidence level is generally described as _____, where C represents the area between two z -scores we call _____.
- A critical value is the z -score that sets the boundary for our confidence interval (how many standard deviations on either side do we want to go).
- It is labeled as _____

Calculating z^*

On TI-84:

2^{nd} $\rightarrow$ Vars $\rightarrow$ 3:invNorm

- area: represents the proportion in the area you are trying to find
- standard normal distribution: μ is always 0 and σ is always 1 when finding critical values
- tail: where is the area you entered above located?



Find z^* for the following confidence levels.

80% Confidence:

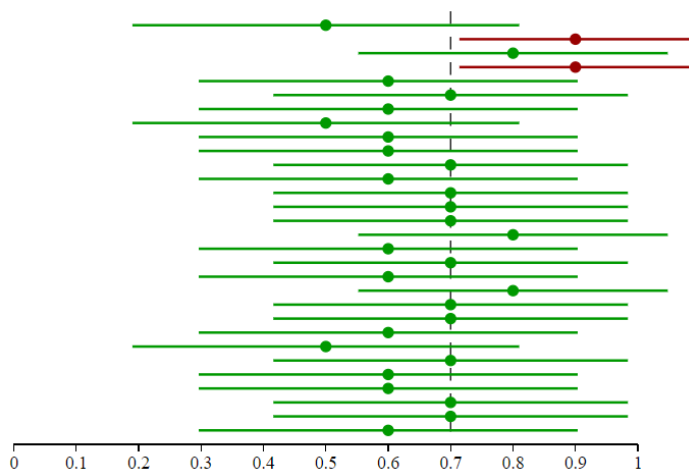
90% Confidence:

95% Confidence:

99% Confidence:

Long-Term Capture Rate

- A C% confidence interval gives an interval of _____ for a parameter
- The interval is calculated from the _____
- The confidence level C gives the _____ rate for calculating the confidence interval. That is, in C% of all possible samples, the method would yield an interval that captures the true parameter value.
- In other words, if we repeat this process many times and calculate a confidence interval each time, C% of them will capture the true parameter.



If we were to repeat our opening activity every day, we would get various confidence intervals due to the changing value of $\hat{p}$

Our 95% confidence means that in the long run, we would expect 95% of the intervals we create to correctly capture the population parameter $p = 0.7$

Interpreting Confidence Intervals and Levels

<i>Confidence Intervals</i>	“We are C% confident that the interval from _____ to _____ captures the [parameter in context]”	“We are 95% confident that the interval from 0.4 to 0.98 captures the proportion of check marks in our population”
<i>Confidence Levels</i>	“If we take many samples of the same size from this population, about _____% of them will result in an interval that captures the actual parameter value.”	“If we take many samples of size 10 from the population of 100 cards, about 95% of them will result in an interval that captures the true proportion of check marks in our population.”

- On a given problem, you may be asked to interpret the confidence interval, confidence level, or both.
- Be sure you understand the difference: the confidence interval gives a set of plausible values for the parameter and the confidence level describes the long-run capture rate of the method.

Conditions for a Confidence Interval

- A confidence interval is constructed with sample data, so the distribution we will use is the sampling distribution for a sample proportion.
- Therefore, the same conditions we had for sampling distributions for sample proportions, we have for confidence intervals!

<i>Random</i>	<i>Independent</i>	<i>Normal</i>

- When calculating our standard deviation of our sampling distribution, because we do not know “p” from our population, we replace p with $\hat{p}$.
- The resulting quantity is known as the _____ of the sample proportion $\hat{p}$.
- This describes how close the sample proportion $\hat{p}$ will typically be to the population proportion p in repeated SRSs of size n.

Calculating a Confidence Interval

When conditions are met, a C% confidence interval for the unknown proportion p is:

where z^* is the critical value for the standard normal curve with C% of its area between $-z^*$ and z^*

This is called a _____

Example: Constructing a Confidence Interval for a Proportion

A recent survey wanted to see what proportion of high school students believed there is life on Mars. The poll’s population was all American high school students. They selected an SRS of 430 students and found that 204 of them believe that there is life on Mars. Construct and interpret a 96% confidence interval for the proportion of high school students who believe there is life on Mars.

Step 1	<i>Communicate:</i> Communicate the parameter p in the context of the problem and write down the confidence level you are using	
Step 2	<i>Conditions:</i> Check the conditions required (random, independent, normal)	

<i>Step 3</i>	Calculate: Identify the inference method and construct the confidence interval by writing down the formula, the values substituted into the formula, and then the interval itself.	
<i>Step 4</i>	Conclude: Interpret the interval in the context of the problem using the script.	